

Education

The University of Texas at Austin (GPA: 3.8/4.0)

May 2027

B.S. Electrical Engineering, B.S. Physics, B.S. Mathematics, Certificate German

Relevant Coursework: Fluid Dynamics, Mechanics of Solids, Materials Engineering, Thermodynamics, Dynamics

Graduate Coursework: Magnetic Materials and Devices, Computational Physics, Quantum Mechanics I

Experience

Emerson – Electrical Engineering Design Intern

May 2026–Aug 2026

- Authored configuration and assembly drawings from scratch for the LT-1 transmitter, documenting every valid mounting and assembly variant
- Captured a 4–20 mA op-amp and potentiometer signal-conditioning schematic enabling field calibration of set-points of a reverse-engineered raw production PCB
- Collaborated with Production, Manufacturing, and Engineering teams to author product Work Instructions

Adom Industries Inc. – R&D Design Engineer Intern

May 2025–Aug 2025

- Optimized chassis topology through iterative FEA, cutting material usage 88% while preserving structural integrity for payload transfer
- Architected a 48V VCU (RP5+RP2040) for ROS path planning via CAN and a hardware E-stop and isolated 5V logic rail
- Authored Design for Manufacturing (DFM) docs covering wiring looms, thermal interfaces, and EMI shielding

Sandia National Laboratories – R&D Computational Engineer

May 2024–Aug 2024

- Modeled radial Beryllium liner implosions with ODE/trajectory solvers, comparing ALEGRA predictions against observed fluid instability spectra (RTIs/RMIs)

Projects and Organizations

FSAE Longhorn Racing – Dynamics + Vehicle Modeling Engineer, Steering Lead

Sep 2022–Aug 2025

- Owned, designed, and manufactured the steering system from steel to a carbon-fiber layup with a metal 3D-printed bevel gearbox, cutting weight 68% at equal torsional rigidity
- Authored precision manufacturing drawings for the steering column and related mechanical assemblies, applying GD&T standards to manage tolerance stack-ups and ensure seamless physical integration
- Tuned anti-Ackermann geometry, roll center, and bump steer in a Model-in-the-Loop simulation, validating tire and slip-angle targets against track telemetry

Racing Simulator – Pedal Box & Steering Assembly

- Designed a fully constrained spring-actuated pedal box with driver-adjustable travel and a coaxial rotary-encoder mount for displacement sensing
- Modeled a table-mounted gyroscope-integrated steering assembly for driver use and PCB mounting

UAV Airframe & Flight Controller

- Modeled the rigid-body flight dynamics about hover to size motor/propeller actuation and shape the closed-loop attitude response of a custom quadrotor airframe
- Designing a custom flight controller mounting system utilizing vibration dampeners, successfully isolating sensitive on-board IMUs from high-frequency ESC and rotor vibrations to ensure stable flight dynamics

High-Power E-Bike Boost Converter

- Engineered the mechanical packaging for a high-power DC-DC boost converter, sizing and placing custom heatsinks to optimize thermal management and ensure reliable heat dissipation under peak loads

Banerjee Lab – 2D Material Synthesis & Characterization

Sep 2025–Present

- Grew and characterized MoS₂ monolayers via XRD, Raman, and Photoluminescence spectroscopy

Skills

Languages: C, C++, ARM, Rust, Python, MATLAB, Bash, JS, HTML, CSS, LaTeX

Tools: SOLIDWORKS, Fusion360, Ansys, KiCAD, Cadence, Git/GitHub, Draftsight, LTSpice, MuMax3